HTML Day 2 — Forms, Tables & Semantic Elements

Detailed theory notes for internship learning and revision.

1. HTML Tables

• Tables display data in rows and columns.

• **<table>** creates the table; **<tr>** creates a row; **<th>** creates a heading cell; **<td>** creates a data cell.

• **<thead>**, **<tbody>**, and **<tfoot>** group table header, body, and footer content.

• **colspan** lets a cell span columns; **rowspan** lets a cell span rows.

2. HTML Forms

• Forms collect user information such as login details, registration data, feedback, and contact information.

• **<form>** defines a form. **action** specifies where submitted data is sent. **method** specifies how it is submitted.

• **GET** commonly sends data in the URL and is useful for searches. **POST** sends data in the request body and is commonly used for submitting or creating data. POST alone does not encrypt data; HTTPS provides transport encryption.

3. Input Types

• The **<input>** element supports many controls: **text**, **email**, **password**, **number**, **date**, **radio**, **checkbox**, **file**, **submit**, and **reset**.

• Radio buttons are generally used to select one option from a group; checkboxes allow one or more selections.

4. Labels

• **<label>** provides a description for a form control and improves usability and accessibility.

• A label can be connected to an input through the label's **for** attribute and the input's matching **id**.

5. Buttons

• **<button>** creates a clickable button. It can submit a form, reset it, or perform another action.

6. Dropdowns

• **<select>** creates a dropdown and **<option>** defines each available choice.

7. Text Areas

• **<textarea>** provides a multi-line text input area. It is useful for messages, feedback, comments, descriptions, and addresses.

8. Semantic HTML

• Semantic HTML uses elements that clearly communicate the meaning and structural purpose of content.

• Semantic elements help developers, browsers, search engines, and accessibility tools understand page structure.

9. Semantic Elements

• **<header>** — introductory content for a page or section, such as a title or logo.

• **<nav>** — a significant group of navigation links.

• **<main>** — the primary content of the webpage.

• **<section>** — a thematic group of related content, usually with a heading.

• **<article>** — self-contained content that can make sense independently, such as a blog post or review.

• **<footer>** — closing information such as copyright, contact details, or related links.

10. Semantic Page Structure

• A common conceptual structure is: **header** → **nav** → **main** → **section/article** → **footer**.

• Semantic elements communicate meaning and structure; they do not automatically provide visual styling. CSS controls appearance.

11. Quick Revision

• **table / tr / th / td** = table / row / heading cell / data cell.

• **form** = collect user input.

• **label** = describe a form control.

• **select / option** = dropdown / choice.

• **textarea** = multi-line input.

• **header / nav / main / section / article / footer** = semantic page structure.